



Literature Results

Objective

I am developing a real-time closed-loop system for detecting interictal epileptiform discharges, also called IEDs, using an STM32 NUCLEO-H723ZG microcontroller board.

Please conduct a broad exploratory research review to identify the best possible IED detection methods that could realistically run in real time on this specific microcontroller platform.

Important constraint:

Do not assume a specific detection method in advance. Start from the literature and engineering evidence, identify what methods have been used for IED detection or closely related epileptiform-event detection, and then evaluate which of those methods are suitable for implementation on the STM32 NUCLEO-H723ZG.

The main goal is to improve IED detection on my board while maintaining strong sensitivity and specificity, low false-positive rate, low latency, and reliable real-time operation.

Please focus on methods that are suitable for:

- * Closed-loop stimulation/control
- * Real-time causal detection
- * Embedded microcontroller implementation
- * Intracranial electrophysiology, local field potential, hippocampal recordings, thalamic recordings, or similar neural signals
- * Experimental epilepsy models, including kainic acid, kindling, or related seizure/IED models

For each method you find, explain:

1. What signal features or principles the method uses
2. Whether it was designed for IED detection, seizure detection, spike detection, or another related task
3. Whether it is suitable for real-time closed-loop use
4. Whether it can realistically run on an STM32 NUCLEO-H723ZG
5. Computational complexity
6. Memory requirements
7. Detection latency
8. Reported sensitivity, specificity, precision, false-positive rate, or related performance metrics
9. Robustness to noise, baseline drift, stimulation artifact, movement artifact, and amplitude variability
10. Ease of implementation in embedded C/C++

11. Whether it needs floating point, fixed point, large buffers, training data, or offline preprocessing
12. Strengths and weaknesses
13. Key papers or systems that used this type of method

Please specifically search for:

- * IED detection methods in neuroscience and epilepsy literature
- * Real-time epileptiform discharge detection
- * Closed-loop epilepsy detection systems
- * Embedded seizure or IED detection systems
- * Microcontroller-based neural event detection
- * Low-power or implantable seizure/IED detection algorithms
- * Algorithms used in responsive neurostimulation or closed-loop stimulation
- * Methods validated on intracranial EEG, LFP, hippocampal recordings, or animal epilepsy models

Please separate findings into categories:

1. Methods clearly suitable for STM32 NUCLEO-H723ZG implementation
2. Methods possibly suitable after simplification
3. Methods likely too computationally expensive
4. Methods that are accurate but mostly offline
5. Methods that are promising but lack enough validation

Then rank the top candidate methods for my use case.

For the final recommendation, provide:

1. The most practical detection approach for STM32 NUCLEO-H723ZG
2. A backup approach if the first one has too many false positives
3. A more advanced approach if additional computational budget is available
4. Suggested sampling rates
5. Suggested preprocessing/filtering strategy
6. Suggested feature windows in milliseconds
7. Suggested adaptive threshold strategy
8. Suggested artifact rejection strategy
9. Suggested refractory/lockout period
10. Suggested performance metrics for validation
11. A plan for tuning the detector using manually labeled IEDs
12. A practical implementation roadmap from offline Python validation to embedded STM32 deployment

Please cite primary papers, review papers, and engineering examples wherever possible. Clearly distinguish evidence from IED-specific studies versus seizure-detection studies. Also clearly distinguish offline algorithms from algorithms that are genuinely appropriate for real-time closed-loop control.

Results

Objective

Identify interictal epileptiform discharge (IED) / interictal spike detection approaches from the literature and evaluate which can realistically run as *causal*, *real-time*, *closed-loop* detectors on an STM32 NUCLEO-H723ZG (STM32H723, Cortex-M7 up to 550 MHz), prioritizing low latency, low false positives, and implementability in embedded C/C++. Evidence is distinguished between IED-specific detectors and seizure detectors that can be adapted.

Platform constraints (STM32H723ZG class)

The STM32H723ZG is a Cortex-M7 class MCU reported operating at 550 MHz, with DMA-capable high-throughput peripherals and strong sensitivity of latency to cache configuration; a real-time DMA/peripheral study on the STM32H723ZG class measured ~792–950 Mb/s transfers (PSSI example) and shows that real-time streaming + processing is feasible when DMA and release optimizations are used. ([1.1], [1.2])

Implication: algorithms that are $O(N)$ per sample with small state (thresholding, line length, half-wave state machines) are trivially feasible; moderate transforms (small DWT or short FFT/bandpower) are feasible; heavy models (large CNN/LSTM) may be feasible only if aggressively quantized and designed around short windows.

Candidate method families (exploratory review)

A comparative summary table is provided below.

Method family	Primary features / principle	Typical window / latency	Reported performance metrics	Compute / memory notes	Robustness notes	Embedded feasibility on STM32H723	Key citations
Normalized threshold (IED-specific)	Baseline flattening / moving-average normalization, then scalar threshold on normalized deviation to mark brief spikes	Suited to sub-250 ms spikes; effectively sample-by-sample or very short sliding window; MATLAB runtime suggests real-time margin	Human iEEG IIS detector: average TPR >98%, FPR <2%; consolidated TPR ~98.8%, FPR ~2%; one hour processed in ~13 s in MATLAB ([2.1], [2.2])	Very low compute; one-pass arithmetic plus short baseline estimate; tiny state, tiny RAM; no training beyond threshold selection	Good when morphology is sharp and channel SNR is high; sensitive to threshold drift, artifacts, stimulation transients, amplitude variability unless adaptive baseline and lockout are added	High	([2.1], [2.2])
Line length / area / energy / slope adaptive threshold	Time-domain features such as line length, area/integrated envelope, amplitude, slope, energy; adaptive threshold from recent feature statistics	Short windows; one hardware design reports detection every 64 ms; RNS-style line-length and area are intended for early responsive detection ([3.1], [4.1], [5.1])	Closed-loop seizure chip: >90% accuracy, 64 ms detection interval (not IED-specific) ([3.1]); cloud-gating work found 2 band-power features often beat adding line length, but line length remained device-compatible ([6.1], [6.2])	Extremely light; one design stores only 4 recent feature values for adaptive thresholding; line length is sum of abs first differences; area is simple accumulation; tiny RAM, integer-friendly ([3.2], [7.1], [4.1])	Fairly robust to baseline drift if based on derivatives / abs differences; vulnerable to movement and stimulation artifact because artifact inflates all time-domain features; needs blanking/refractory	High	([3.2], [3.1], [7.1], [4.1], [5.1])
Half-wave / morphology finite-state	Spike morphology using amplitude, duration, slope, curvature, half-wave measures; can be encoded as deterministic finite automata / rule sets	Usually causal over tens of ms to ~100 ms half-wave duration; low intrinsic algorithmic latency	Review cites DFA spike using amplitude+slope with 99.13% recognition; Gotman-style half-wave methods reported 0.79 false positives/min in reviewed literature ([8.1])	Low compute; comparisons, timers, slope estimates, simple state machine; small buffers only	Better than pure thresholding for shape discrimination and suppressing slow artifacts; still sensitive to stimulation artifact and waveform variability across preparations unless thresholds are adaptive	High	([8.1])

Method family	Primary features / principle	Typical window / latency	Reported performance metrics	Compute / memory notes	Robustness notes	Embedded feasibility on STM32H723	Key citations
Bandpower + logistic regression / linear classifier	Welch PSD or short spectral bandpower in selected bands; optionally a few correlation features; linear classifier used as hypersensitive gate	Typically 1 s clips with moving accumulation; algorithmic latency <10 ms for 1 s clips in hybrid system; local off-board total latency <200 ms in trialed setup ([6.3], [6.4], [6.5])	Hybrid gate+cloud system: 99% seizure sensitivity overall, FPR 0.03/h; gate alone: single-feature >=92% per subject, 95% overall; five-feature >=98% per subject, 99% overall; 22.7x data reduction ([6.3], [6.2], [6.6])	Light to moderate; a few spectral features plus logistic regression; authors note performance plateaued after 2-3 features, making analog/on-chip implementation feasible ([6.3], [6.4])	Spectral features can be more robust to isolated sharp noise than raw thresholding, but require enough window length; less ideal for very short IEDs unless window shortened and bands tuned; can miss morphology-specific spikes	Medium-High	[[6.1], [6.3], [6.4], [6.5], [6.2], [6.7]]
Wavelet (DWT/CWT/SWT)	Multiscale transient enhancement; wavelet coefficients thresholded for spikes/discharges	Often tens to hundreds of ms windows; can be real-time but latency depends on transform frame and mother wavelet	Real-time rodent spike-wave discharge detector based on CWT reported feasible real-time operation (paper identified, detailed metrics not extracted here) ([9.1]); review notes wavelet-domain spike detectors are common ([8.1])	Higher compute than time-domain methods; memory for co-efficient buffers; DWT moderate, CWT heavier; fixed-point implementation possible but more effort	Strong at transient/noise separation and amplitude variability; better than raw thresholds for low-SNR spikes; can still be disturbed by stimulation artifact unless blanked	Medium	[[9.1], [8.1]]
AR model / autoregressive spectral change	Online AR parameter estimation or prediction-error change to detect abnormal dynamics	Short recursive windows; potentially low latency if updated sample-by-sample	Search identified implantable AR-model seizure detector literature, but concrete extracted sensitivity / FPR metrics were not available in current context	Moderate compute: recursive coefficient update, multiply-accumulates, stability handling; modest memory for taps/state	Can normalize amplitude effects and track rhythmic changes; less directly matched to isolated sharp IED morphology than threshold/morphology detectors	Medium	[[9.1]]
Time-series segmentation (TSS: Zdensity RODE / AMPDE)	Segment signal into events using adaptive z-score or scalogram density; classify baseline vs interictal vs ictal from event amplitude range / duration and density structure	Validated at 2 kHz LFP; causal event-oriented processing; latency not numerically reported	4-AP mouse hippocampus-cortex slices: ZdensityRODE ictal P/R 93/93%, interictal precision 42%; AMPDE ictal precision/recall 96/90%, interictal precision 54% ([10.1], [10.2])	Designed explicitly for low-power detection; uses low-cost features (amplitude range, duration); AMPDE higher cost than ZdensityRODE; no op counts reported ([10.2], [10.3])	Correlates with SNR; promising for LFP event parsing, but current interictal precision is modest and validation is mainly in vitro rodent slices	Medium	[[10.1], [10.2], [10.3]]
Tiny CNN / compact DL	Small 1D/2D CNN on raw EEG or compact spectrogram; optional majority voting	Usually 1-10 s windows in seizure studies; latency from window length dominates; examples: 1 s stride, median delays 3.7 s for seizure detection, 2 s windows updated every 100 ms in one system ([11.1], [12.1])	SeizureNet on implantable MCU: sensitivity 0.96, FP/h 10.1, median delay 3.7 s ([11.2]); edge MCU DL: CNN 96.70% / 0.102 h ⁻¹ , LSTM 97.61% / 0.071 h ⁻¹ for seizure event detection ([13.1], [13.2])	Much heavier than feature rules but feasible on MCU after compression/quantization; example CNN ~2.4M MACs per 256x9 input; quantization cuts memory/power with <0.5% accuracy loss; memory may still dominate ([13.1], [13.2], [13.3])	Potentially most tolerant to morphology variability and complex noise if trained well; but needs labeled data, careful generalization checks, artifact examples, and usually longer windows than ideal for fast IED-triggered stimulation	Medium	[[13.1], [11.1], [11.2], [12.1], [13.2], [13.3]]
Tiny LSTM / RNN	Sequential modeling of raw signal dynamics; event decision via recurrent state and voting	Similar to CNN but often higher latency/power; seizure systems used sliding windows and WMV	Edge DL study: LSTM 97.61% sensitivity, 0.071 h ⁻¹ FPR, best among tested seizure detectors ([13.1], [13.2])	Highest compute/power among tested edge DL models; more memory traffic; harder to implement and debug than CNN or rules ([13.1])	Can capture temporal context and suppress isolated false alarms, but less attractive for tight closed-loop MCU budgets and very short IEDs	Medium-Low	[[13.1], [13.2]]

Method family	Primary features / principle	Typical window / latency	Reported performance metrics	Compute / memory notes	Robustness notes	Embedded feasibility on STM32H723	Key citations
Cloud-gating hybrid	Ultra-light on-device gate sends candidates to heavier remote classifier; on-device gate often bandpower / line length + linear model	1 s candidate clips in cited system; end-to-end algorithmic latency <10 ms for clip analysis; local off-board total latency <200 ms ([6.3], [6.4])	99% seizure sensitivity, 0.03/h FPR, all transmitted seizures detected by cloud stage ([6.3], [6.6])	On-device cost low; remote stage can be very heavy (e.g., 3000-tree RF). Not suitable if stimulation must remain fully autonomous and communication-free	Strong if connectivity is reliable; poor fit for standalone microcontroller-only closed-loop stimulation because telemetry gaps and cybersecurity become system-level risks	Low for standalone STM32-only closed loop; High as development/validation architecture	([6.3], [6.4], [6.5], [6.2], [6.6], [6.7])

Table: This table compares major candidate method families for real-time epileptiform event detection, emphasizing what has direct evidence for embedded or closed-loop use and how suitable each family is for an STM32H723-class implementation.

Method family A — Normalized threshold IED detector (IED-specific)

1. Features/principle: baseline flattening via moving-average normalization, then threshold on normalized deviations to mark spikes. ([2.1], [2.2]) **2. Task:** explicitly interictal spike/IIS/IED detection on human intracranial EEG. ([2.2]) **3. Closed-loop suitability:** yes (causal; decisions can be made sample-by-sample or with a short confirmation rule). ([2.1]) **4. STM32H723 feasibility:** high; O1 arithmetic per sample plus a lightweight baseline estimator. ([2.1]) **5. Complexity:** ~few adds/mults/comparisons per sample (+ optional running mean/median/MAD/SD). ([2.1]) **6. Memory:** tiny (baseline state + event bookkeeping). ([2.1]) **7. Latency:** fundamentally tens of ms (spike duration) + optional confirmation window; not explicitly reported in the paper. ([2.2]) **8. Performance metrics:** average true positive rate >98% and false positive rate <2% across 7 iEEG patients; consolidated TPR ~98.8% and FPR ~2% reported. ([2.1], [2.2]) **9. Robustness:** depends strongly on normalization and artifact handling; threshold detectors are sensitive to stimulation artifact and baseline changes unless adaptive thresholds and lockouts are used (not fully quantified in the excerpt). ([2.1]) **10. Embedded implementation:** easy in C; fixed-point is straightforward. **11. Numeric requirements:** does not require floating point; can be fixed-point; needs a baseline window estimate but not training data (beyond threshold tuning). ([2.1]) **12. Strengths/weaknesses:** strongest for very low latency and simplicity; weakness is false positives from artifacts and morphology variability unless paired with additional morphology rules. **13. Key paper:** Palepu et al. 2018. ([2.1], [2.2])

Method family B — RNS-style time-domain detectors (line length, area, bandpass; plus simple adaptive thresholds)

1. Features/principle: line length (sum of absolute first differences), area under envelope/integral, bandpass/half-wave detectors; simple thresholds. ([4.1], [5.1], [7.1]) **2. Task:** primarily seizure onset detection in implantable responsive neurostimulation; not IED-specific, but the same primitives are appropriate for sharp transients. ([5.1], [4.1]) **3. Closed-loop suitability:** yes; these detectors are explicitly used in chronic responsive stimulation systems. ([5.1]) **4. STM32H723 feasibility:** high; all features are O(N) with small constants and minimal buffering. ([7.1], [4.1]) **5. Complexity:** per sample: abs/sub for line length; accumulate for area/energy; optional running statistics. ([7.1]) **6. Memory:** tiny (window accumulators). ([7.1]) **7. Latency:** depends on chosen window; an embedded seizure chip example performs detection every 64 ms (not IED-specific). ([3.1]) **8.**

Performance metrics: RNS papers in this context provide detector *types* and acquisition parameters (e.g., 250 Hz sampling; 4–90 Hz bandpass) but not IED-specific sensitivity/FPR in the extracted excerpts. ([4.1], [5.1]) **9. Robustness:** line length/derivative features reduce baseline drift sensitivity, but stimulation/movement artifacts inflate all time-domain features; blanking and refractory are typically required. ([4.1]) **10. Embedded implementation:** very easy in C; fixed-point friendly. **11. Numeric requirements:** fixed-point feasible; no training required. **12. Strengths/weaknesses:** excellent embedded suitability and low latency; false positives likely without additional morphology gating and artifact mitigation. **13. Key evidence:** RNS detector families described (bandpass, line length, area; plus noise/saturation detectors). ([5.1])

Method family C — Half-wave / morphology rule systems (finite-state style)

1. Features/principle: amplitude, duration, slope/sharpness/curvature and half-wave measures; can be encoded as deterministic finite-state logic (e.g., “DFAspike”). ([8.1]) **2. Task:** spike detection (largely EEG/MEG literature; adaptable to iEEG/LFP). ([8.1]) **3. Closed-loop suitability:** yes (causal, low-latency). **4. STM32H723 feasibility:** high. **5. Complexity:** O1 per sample + a few counters and comparisons. ([8.1]) **6. Memory:** tiny. **7. Latency:** tens of ms. **8. Performance metrics:** review cites DFAspike (amplitude+slope) with 99.13% recognition rate; Gotman-style half-wave methods with ~0.79 false positives/min after state classification (contextual review evidence; not necessarily intracranial). ([8.1]) **9. Robustness:** improves over pure thresholding by enforcing shape constraints; still sensitive to stimulation artifact unless blanked. **10–13. Implementation:** straightforward C; fixed-point; no training.

Method family D — Spectral bandpower + linear classifier (adaptable; evidence from implantable “gating”)

1. Features/principle: short-window bandpower features (Welch PSD over selected bands), optionally cross-correlation and line length; logistic regression classifier. ([6.5], [6.2]) **2. Task:** seizure detection, designed to be device-compatible for implantable systems; can be adapted as a *second-stage false-positive reducer* for IED candidates. ([6.4], [6.5]) **3. Closed-loop suitability:** yes (causal), but note the cited system used cloud for final precision; the *gate* itself is embedded-feasible. ([6.4]) **4. STM32H723 feasibility:** medium-high (few FFT/bandpower computations per update). **5. Complexity:** moderate; depends on FFT length and update cadence. **6. Memory:** small-moderate (FFT buffers). **7. Latency:** the cited pipeline operating on 1 s clips reports algorithmic latency under 10 ms for clip processing (computation), but detection latency is dominated by windowing/accumulation (seconds in seizure context). ([6.3], [6.2]) **8. Performance metrics (seizure):** combined gate+cloud achieved 99% sensitivity with FPR 0.03/h; the gate alone achieved ≥98% per subject with five features. ([6.3], [6.2]) **9. Robustness:** bandpower can be more robust to isolated spikes/noise bursts than raw amplitude; however, for *IED detection* very short events may be diluted in longer windows unless windows are shortened and bands are chosen for transient energy. ([6.2]) **10–11. Implementation:** feasible in C using CMSIS-DSP; fixed-point FFTs possible. **12. Strengths/weaknesses:** good as an artifact/false-positive filter, less direct for ultra-short IEDs if using long windows. **13. Key paper:** Baldassano et al. 2019 (device-compatible gate). ([6.5], [6.2])

Method family E — Time-series segmentation (TSS) for LFP epileptiform events (promising, limited interictal precision)

1. Features/principle: segment LFP into events and classify based on event amplitude range/duration and density measures; designed for low-power embedded detection. ([10.3]) **2. Task:** epileptiform pattern recognition in vitro (4-AP slices), with both ictal and interictal outputs. ([10.2]) **3. Closed-loop suitability:** potentially (event-based, causal), but closed-loop latency details are not reported. ([10.3]) **4. STM32H723 feasibility:** medium; designed for computationally inexpensive implementations, though AMPDE is higher cost than ZdensityRODE. ([10.3]) **5–7. Complexity/memory/latency:** not quantified as ops/bytes in the excerpt. **8. Performance metrics:** ictal detection precision/recall ~93–96%/90–93%, but interictal precision is modest (42–54%); suggests many

false positives for interictal unless tuned. ([10.1], [10.2]) **9. Robustness:** performance correlated with SNR, implying susceptibility to noise. ([10.1]) **10–13. Implementation:** plausible in C; may need parameter tuning/training per preparation.

Method family F — TinyML / compact deep learning on microcontrollers (advanced; depends on labeled data)

Evidence here is largely seizure detection, but it is instructive for what is feasible on MCUs and what windowing/quantization tradeoffs look like.

Edge DL on MCU: quantized/compressed CNN/LSTM/DNN models ran on a general MCU; CNN required ~2.4 M MACs for a 256×9 input; LSTM had best sensitivity/FPR but highest power/latency; quantization reduced inference time/memory/power without performance loss. ([13.1])

SeizureNet on an implantable MCU: a CNN designed for an ultra-low-power microcontroller achieved sensitivity 0.96 with false detection rate 10.1/h and median detection delay 3.7 s (seizure detection). ([11.2])

Energy/ops comparison: for seizure detectors, a random forest (handcrafted features) required ~67k arithmetic ops and 67k memory accesses per classification, versus ~488k AOs/963k MAs for CNN and ~772k/978k for RNN (platform-independent energy model validated on MCU). ([14.1])

Implication for IED detection on STM32H723: a small, quantized CNN could be feasible as a second-stage classifier on short windows (e.g., 128–256 ms) if trained on labeled IED vs artifact; however, training data requirements and robustness to stimulation artifact must be explicitly addressed.

Categorization by STM32H723ZG suitability

1) Clearly suitable for STM32H723ZG (real-time closed-loop, causal)

- **Normalized threshold IED detector** (Palepu et al.): explicitly IED/IIS; high TPR with low FPR in iEEG; extremely low compute. ([2.1], [2.2])
- **Time-domain features + adaptive thresholds:** amplitude/slope/line length/energy; embedded seizure chip detection every 64 ms suggests low-latency feasibility; aligns with RNS detector families (line length/area/bandpass). ([3.1], [5.1], [7.1])
- **Half-wave / finite-state morphology rules:** low compute and designed for spikes in classic literature. ([8.1])

2) Possibly suitable after simplification

- **Bandpower (few bands) + linear classifier** as a *false-positive filter* or “gate” (shorten windows, reduce band count): embedded-feasible and strong in implantable gating work for seizures. ([6.5], [6.2])
- **Wavelet/DWT-based transient enhancement** (small DWT levels only) if you need more noise robustness than time-domain rules; may require careful optimization. ([8.1])

3) Likely too computationally expensive (for always-on IED detection)

- **LSTM/RNN** as continuous inference: higher compute and memory access burden than RF/CNN in implantable comparisons, and higher power/latency than CNN in MCU DL study. ([14.1], [13.1])
- **Large CNNs / long STFT pipelines** (especially if using long windows); seizure literature shows preprocessing/runtime can dominate. ([11.1])

4) Accurate but mostly offline (or not IED-focused)

- **Cloud-gating architectures:** excellent seizure sensitivity and low FPR, but require reliable telemetry and external compute; not appropriate if stimulation must be fully autonomous on the STM32. ([6.3], [6.4])

5) Promising but lacks enough validation for your target

- **TSS methods (ZdensityRODE/AMPDE):** explicitly motivated by low-power embedded LFP processing, but interictal precision in the cited in vitro validation is modest and latency/embedded footprints are not fully specified. ([10.1], [10.3])

Ranked top candidate methods for STM32H723 closed-loop IED detection

Ranking criterion: (i) causal latency in the 10–100 ms range, (ii) low false positives under artifact, (iii) no heavy training requirements, (iv) easy embedded implementation.

1. Two-stage time-domain detector: candidate threshold + morphology confirmation (half-wave/DFA) + adaptive baseline

- Anchors: normalized threshold IED detector with strong iEEG metrics (Palepu) plus morphology-rule family (review) plus line length/energy features used in implantable detectors. ([2.1], [8.1], [5.1])

2. RNS-like line length/area/energy detector with adaptive thresholds + artifact blanking/refractory

- Anchors: RNS detector families (bandpass/line length/area) and acquisition specs; line length described as computationally efficient. ([5.1], [4.1])

3. Lightweight spectral “sanity check” (few bandpowers) as a second-stage false-positive reducer

- Anchors: device-compatible gating work showing a few bandpower features + logistic regression are sufficient for high sensitivity in seizures; performance plateau after 2–3 features suggests embedded feasibility. ([6.4], [6.2])

4. Quantized tiny CNN on short windows (advanced option)

- Anchors: MCU DL implementations and quantization results; caution that the evidence is seizure-focused and window lengths are longer than typical IEDs. ([13.1], [11.2])

Final recommended detection approaches (actionable)

1) Most practical approach (recommended primary): “Fast candidate + morphology confirmation”

Pipeline (per channel, causal):

1. **Pre-filter:** bandpass to emphasize spike band (commonly 4–90 Hz used in an implantable RNS context; you can adapt for hippocampal LFP and your IED morphology). ([4.1])
2. **Baseline normalization:** moving-average or robust scale estimate to flatten drift (as in Palepu). ([2.1])
3. **Candidate trigger:** threshold on normalized deviation and/or on derivative/line length increment. ([2.1], [7.1])
4. **Morphology confirmation (10–80 ms):** half-wave state machine enforcing (a) sharp slope, (b) duration range (IED typical 20–70 ms in review), (c) post-spike slow-wave optional. ([8.1])
5. **Lockout/refractory:** prevent retriggering and avoid stimulation “echo”.

Why this is best for STM32H723: it directly targets the short transient nature of IEDs with minimal buffering, matches published IED-specific evidence for high TPR/low FPR with low runtime, and adds morphology rules to cut artifact-driven false positives. ([2.1], [8.1])

2) Backup approach (if too many false positives): add a short spectral gate

If artifacts dominate, add a **second-stage bandpower classifier** on a short (e.g., 128–256 ms) window centered on the candidate. Keep 2–3 bands and a linear model (logistic regression), mirroring the “few features suffice” observation in implantable gating. ([6.4], [6.2])

3) More advanced approach (if more compute budget is available): quantized tiny CNN classifier on candidate snippets

Train a small 1D CNN to classify *candidate snippets* (IED vs artifact vs physiological sharp-wave/ripple-like events), using quantization to reduce memory/power as shown in MCU DL work. ([13.1]) This is the highest expected robustness to morphology variability, but requires labeled training data and careful artifact coverage.

Suggested engineering parameters (starting points)

These are starting points consistent with the cited embedded/implant evidence; they must be tuned on your signals.

1. Sampling rates:

- **IED detection:** 500–2000 Hz recommended depending on whether you need to preserve fast slopes and hippocampal sharp transients; a validated LFP slice study used 2 kHz. ([10.3])
- **If you must minimize compute:** 250–500 Hz can work for many iEEG/LFP applications; implant context example used 250 Hz with 4–90 Hz bandpass. ([4.1])

2. Preprocessing/filtering strategy:

- Bandpass resembling 4–90 Hz is a pragmatic baseline for iEEG/LFP closed-loop detectors (implant evidence). ([4.1])
- Add notch only if needed; keep filters causal (IIR biquads) and avoid long FIRs unless necessary.

3. Feature windows (ms):

- Candidate detector: 10–30 ms slope/derivative accumulation.
- Morphology confirmation: 20–100 ms (reviewed spike durations ~20–70 ms). ([8.1])
- Optional slow-wave confirmation: extend to 150–250 ms.

4. Adaptive threshold strategy:

- Maintain a robust running estimate of baseline scale; Palepu uses normalization to reduce dependence on clinician “mental thresholds.” ([2.2])
- If you prefer feature-statistic thresholds, the “mean + std of recent feature values” adaptive threshold idea is a known embedded-friendly design pattern (though shown for seizure detection). ([3.2])

5. Artifact rejection strategy:

- **Stimulation blanking:** ignore samples for a short blanking window after stimulation; then ramp thresholds back.
- **Saturation/noise detectors:** follow RNS practice of including detectors for saturation/noise to disable detection temporarily. ([5.1])
- **Cross-channel coincidence rule:** require IED detection on ≥ 2 nearby channels within a short time (e.g., 10–20 ms) if you have multi-contact depth electrodes.

6. Refractory/lockout period:

- 100–300 ms after an IED trigger is a common practical range to reduce double-counting and artifact retriggering; lengthen during stimulation.

7. **Validation metrics to report:**

- Event-level sensitivity/recall, precision (PPV), false positives per minute/hour, and detection latency distribution (median/IQR). Use the same style as Palepu (TPR/FPR) and seizure detector literature (FPR per hour) for comparability. ([2.2], [13.1])

Practical tuning plan using manually labeled IEDs

1. **Label creation:** manually annotate IED peaks (and artifact classes) on representative sessions.
2. **Offline replication:** implement the exact embedded pipeline in Python (fixed-point emulation if possible).
3. **Threshold sweep:** generate ROC/PR curves and pick operating points prioritizing *low false positives* at acceptable sensitivity (as done in Palepu by selecting an ROC-optimal threshold). ([2.1])
4. **Artifact augmentation:** ensure labels include stimulation artifacts and movement artifacts; tune blanking and morphology rules to eliminate these.
5. **Cross-validation across days:** because implanted baselines drift; recalibrate adaptive thresholds periodically.

Implementation roadmap (offline → STM32)

1. **Prototype in Python:**

- Causal IIR filters; normalization; candidate trigger; morphology confirmation; refractory; logging.

2. **Performance test on recorded hippocampal/thalamic LFP:**

- Report sensitivity/precision/FPR and latency; compare to baseline normalized threshold-only detector. ([2.1])

3. **Port to embedded C/C++ on STM32H723:**

- Use DMA for acquisition and ring buffers (hardware feasibility for high-throughput DMA is demonstrated in H723 studies). ([1.1])
- Use fixed-point where possible; if using bandpower/FFT, use CMSIS-DSP and keep FFT sizes small.

4. **Hardware-in-loop testing:**

- Replay recorded data through DAC/ADC or stream via UART/SPI to validate real-time behavior and latency.

5. **Closed-loop safety features:**

- Saturation/noise detection; stimulation blanking; daily therapy limits and watchdogs (RNS-style design principle). ([5.1])

6. **Optional advanced step:**

- If a tiny CNN is added, quantize to int8 and measure MACs/latency on target; MCU DL work shows quantization preserves performance while reducing memory/power. ([13.1])

Key takeaways

- The strongest *IED-specific* evidence in this corpus supports **simple normalized thresholding** as high-performing and computationally trivial, making it an excellent starting point for STM32 closed-loop IED triggering. ([2.1], [2.2])
- Implantable responsive systems and implant-focused literature strongly support **time-domain features (line length/area/energy) and simple thresholds** as practical embedded primitives, but you should add **morphology constraints + artifact blanking** to control false positives in a stimulation setting. ([4.1], [5.1], [7.1])
- If false positives remain problematic, a **minimal bandpower + linear model “sanity check”** is an embedded-feasible way to reduce false alarms, inspired by device-compatible gating literature (seizure-focused but transferable as a second stage). ([6.4], [6.2])
- Deep learning on MCUs is feasible with quantization and careful design, but evidence here is seizure-focused and window-latency is typically seconds; for IED-triggered stimulation it is best used as a *candidate classifier* on short snippets rather than always-on inference. ([13.1], [11.2])

References

[1.1] Suurinopeuksinen tiedonsiirto ja reaaliaikainen datan prosessointi STM32-mikrokontrollerilla PSSI-väylän avulla. V Pitkänen. 2025. <https://oulurepo.oulu.fi/bitstream/handle/10024/55126/nbnfioulu-202504152637.pdf?sequence=1> Context: "The excerpt references the STM32H7 series and specifically the STM32H723ZG Nucleo/HW ("STM32H723ZG") and points to core and board documentation ("Arm® Cortex®-M7 32-bit 550 MHz MCU Datasheet – production data" and "STM32H723/733 Reference Manual, RM0468, Rev 3."). The work tested STM32-H723ZG with a PSSI (Parallel Synchronous Slave Interface) peripheral ("PSSI (Parallel Synchronous Slave Interface) -väylän") and used DMA for transfer/performance optimizations ("Tiedonsiirron ja suorituskvyn optimointiin hyödynnettiin muun muassa DMA:ta"). Measured PSSI transfer latencies and throughput are given: "PSSI-väylän siirronkestoksi nykyisellä testauskokeonpanolla saavutettiin 332 µs – 276 µs, mikä vastaa 792–950 Mb/s tiedonsiirtonopeutta." Achieved real-time processing rates on the H723 were ~"1529 kuvaa sekunnissa" and on the U575 ~"1143 kuvaa sekunnissa". References also point to the Nucleo board user manual ("STM32H7 Nucleo-144 boards User manual") and the U5xx family core datasheet ("Ultra-low-power Arm® Cortex®-M33 32-bit MCU datasheet – production data"). The excerpt thus supplies confirmed core type/frequency (Cortex-M7, 550 MHz), presence of PSSI peripheral and DMA use, measured PSSI latency/throughput and observed application-level FPS, but lacks explicit RAM/Flash sizes and detailed peripheral counts — those would be in the cited RM0468 and device datasheet references."

[1.2] Suurinopeuksinen tiedonsiirto ja reaaliaikainen datan prosessointi STM32-mikrokontrollerilla PSSI-väylän avulla. V Pitkänen. 2025. <https://oulurepo.oulu.fi/bitstream/handle/10024/55126/nbnfioulu-202504152637.pdf?sequence=1> Context: "Relevant concrete hardware-related observations for the STM32H723 (Nucleo-H723ZG context) from the excerpt: the H723 was tested with a processor clock of 550 MHz in at least one performance measurement (summation test). The on-chip cache for the H723 was disabled by default (unlike the U575 where cache was enabled by default) but could be activated via software; enabling cache dramatically reduced processing latency (with caveats of data errors in some tests). The PSSI interface (sensor bus) was exercised up to 50–60 MHz without data errors in the test setup; corresponding measured aggregate transfer rates for the H723 reached up to ~950 Mbit/s at 60 MHz (theoretical ~960 Mbit/s). Power was measured via the microcontroller IDD pin; reported H723 currents were on the order of ~94–100 mA across tested PSSI frequencies (power ~320 mW). The software build configuration strongly affected runtime: Release builds used -Ofast in STM32CubeIDE, Debug used -O0, impacting measured latencies. The excerpt does not state core type (Cortex revision), on-chip RAM/ROM sizes, or detailed peripheral counts—those are not provided in this document."

[2.1] Automating Interictal Spike Detection: Revisiting A Simple Threshold Rule. Anil Palepu, Christine Sharmini Premanathan, F. Azhar, M. Vendrame, T. Loddenkemper, C. Reinsberger, Gabriel Kreiman, Kimberly A. Parkerson, S. Sarma, W. Anderson. 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (2018). <https://doi.org/10.1109/embc.2018.8512244> Context: "This paper evaluates a simple threshold-based interictal spike (IIS/IED) detector applied to intracranial EEG from seven patients. Preprocessing includes a moving-average normalization that flattens baseline while preserving spikes. Detection uses a scalar threshold chosen by minimizing Euclidean distance to the ideal ROC point (TPR=1, FPR=0); the optimal threshold found was -0.46 (i.e., mark deviations >0.46 from normalized baseline as spikes). Data were split 50/50 for training/testing. Results: consolidated TPR ~98.8% and FPR ~2% (per-patient TPRs/FPRs shown in Table I, e.g., m10: TPR .991 FPR .008). Computational cost is low: one pass over one hour of data takes ~13 seconds in MATLAB, indicating suitability for real-time/embedded use. Implementation details: events recorded to a 2D array (channels x time) denoted X; evaluation used ROC curves and weighted averaging across patients. Limitations: tested on human intracranial EEG (seven patients, varying electrode locations) so animal/LFP-specific performance and latency on embedded hardware are not reported; further robustness and low-level implementation details (fixed-point arithmetic, memory footprint, per-sample latency) would need characterization for closed-loop embedded deployment."

[2.2] Automating Interictal Spike Detection: Revisiting A Simple Threshold Rule. Anil Palepu, Christine Sharmini Premanathan, F. Azhar, M. Vendrame, T. Loddenkemper, C. Reinsberger, Gabriel Kreiman, Kimberly A. Parkerson, S. Sarma, W. Anderson. 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (2018). <https://doi.org/10.1109/embc.2018.8512244> Context: "This paper evaluates a simple automated interictal spike (IIS) detector based on a threshold rule combined with normalization techniques applied to intracranial EEG (iEEG). The study used iEEG from 7 epilepsy patients (1339 marked IISs across 68 active electrodes) and reports high performance: average true positive rate >98% and false positive rate <2%. Spikes are brief (<250 ms), favoring high temporal-resolution methods. The authors contrast the threshold approach with prior methods: early threshold detectors had poor sensitivity (~63.4%), template-matching can reach ~96% sensitivity but requires clinician-marked examples, and a neural-network/dimensionality-reduction method (Tarassenko et al.) reported 87% sensitivity and 96% specificity but much higher processing time (over 20 minutes vs ~10 seconds for a threshold detector). The paper highlights iEEG's higher SNR and spatial resolution and notes annotations were done by a single clinician on one hour per patient. Overall, the excerpt supports that a normalized threshold detector is low-complexity, fast, and achieves high accuracy on human iEEG, making it promising for real-time/embedded closed-loop use, while template-matching and neural-network approaches trade higher complexity or preprocessing requirements for comparable performance."

[3.1] A Novel Real-Time Threshold Algorithm for Closed-Loop Epilepsy Detection and Stimulation System. Liang-Hung Wang, Zhen-Nan Zhang, Chao-Xin Xie, Hao Jiang, Tao Yang, Qi-Peng Ran, Ming-Hui Fan, I-Chun Kuo, Zne-Jung Lee, Jian-Bo Chen, Tsung-Yi Chen, Shih-Lun Chen, Patricia Angela R. Abu. Sensors (Basel, Switzerland) (2024). <https://doi.org/10.3390/s25010033> Context: "This paper proposes a real-time closed-loop epilepsy detection/stimulation system implemented for on-chip/embedded use. The detection algorithm is time-domain and uses four lightweight features: amplitude, slope, line length, and signal energy. Thresholding is adaptive: thresholds are updated using the mean and standard deviation of four consecutive eigenvalues (parameter combination), enabling continual on-chip updating. The implemented system performs detection every 64 ms and reports >90% accuracy. It was physically designed in UMC 55 nm and verified on an FPGA, and the authors emphasize small area and low power suitable for implanted/intrusive devices. Upon detection the system triggers a two-phase pulse stimulation (34 μ A, 200 Hz). The excerpt addresses seizure detection broadly; it does not explicitly state validation on intracranial EEG/LFP or animal-model interictal spike (IED) datasets nor provide spike-specific sensitivity/false-positive rates, computational op counts, memory use, or runtime complexity. However, the chosen time-domain features and short 64 ms detection interval suggest low-latency, low-complexity methods that are suitable for real-time embedded closed-loop applications."

[3.2] A Novel Real-Time Threshold Algorithm for Closed-Loop Epilepsy Detection and Stimulation System. Liang-Hung Wang, Zhen-Nan Zhang, Chao-Xin Xie, Hao Jiang, Tao Yang, Qi-Peng Ran, Ming-Hui Fan, I-Chun Kuo, Zne-Jung Lee, Jian-Bo Chen, Tsung-Yi Chen, Shih-Lun Chen, Patricia Angela R. Abu. Sensors (Basel, Switzerland) (2024). <https://doi.org/10.3390/s25010033> Context: "This excerpt describes a low-complexity, time-domain threshold algorithm intended for real-time closed-loop epilepsy detection and stimulation. It extracts four simple features: amplitude (AMP), slope (SLP), line length (LL), and energy (POWER), computed from EEG after front-end conditioning (LNA + LPF with 100 Hz cutoff) and 14-bit SAR ADC conversion. Thresholding is implemented via a 4-entry FIFO of recent feature (eigen) values (FV1–FV4); the algorithm computes the mean (AVE) and standard deviation (STD) of those four values and forms a dynamic threshold as $M \times \text{AVE} + N \times \text{STD}$, with eigenvalues and threshold updated synchronously. The paper highlights small area and low power after hardware design and emphasizes ease of implementation, making it suitable for embedded closed-loop hardware. The excerpt does not provide explicit latency numbers, cycle counts, computational complexity in operations per sample, or quantitative detection performance (sensitivity, specificity, false positive rate), nor does it report validation on intracranial EEG/LFP or animal-model datasets. The described method's very small memory (4-value FIFO) and simple arithmetic (mean, std, linear combination) imply low latency and low computational cost, favorable for real-time embedded/implantable systems."

[4.1] Daily Variation in an Intracranial EEG Feature in Humans Detected by a Responsive Neurostimulator System. Robert B. Duckrow, Thomas K. Tcheng. *Epilepsia* (2007). <https://doi.org/10.1111/j.1528-1167.2007.01091.x> Context: "The excerpt describes RNS (eRNS) detection algorithms and implementation details relevant to detection feature design. The device implements three on-board signal-processing detectors: a half-wave detector that can be tuned to a target seizure-onset frequency, a second algorithm that "integrates the area" under the icEEG (description truncated in the excerpt), and a third (not fully described here). A specific feature analyzed was the "line length baseline," described as "a computationally efficient measure of intracranial EEG energy or complexity" that estimates nonlinear energy/complexity of the icEEG and was calculated/reported by the external RNS. Recording/implementation parameters shown: signals bandpass filtered (−3 dB) from 4–90 Hz, recorded by four differential amplifiers from electrode pairs, digitized at 250 Hz with 10-bit precision and sensitivities of 1.17 $\mu\text{V/bit}$ or 0.78 $\mu\text{V/bit}$. The system is intended for chronic closed-loop detection and stimulation to "detect seizures as they begin and disrupt seizure progression." The excerpt provides no explicit numeric latency or detection-timing performance metrics beyond stating the line-length measure is "computationally efficient," and the area-algorithm description is cut off."

[5.1] A Rational Approach to Understanding and Evaluating Responsive Neurostimulation. Nathaniel D. Sisterson, Thomas A. Wozny, Vasileios Kokkinos, Anto Bagic, Alexandra P. Urban, R. Mark Richardson. *Neuroinformatics* (2020). <https://doi.org/10.1007/s12021-019-09446-7> Context: "The excerpt identifies the RNS (NeuroPace) detection algorithm types and some implementation/configuration structure but contains no low-level algorithmic or latency metrics. It states the RNS parameter space splits into detection parameters and stimulation parameters, and that the three primary detector types implemented are bandpass, line length, and area, with additional built-in detectors for saturation and noise. Configuration is per two detection channels (two detecting electrode pairs) and supports up to two first-order Patterns (Pattern A and Pattern B), each optionally composed of up to two second-order patterns (A1, A2, B1, B2). The paper emphasizes that device response depends on physician-selected detection settings matched to patient-specific ECoG seizure onset patterns, and that clinicians observe baseline detector performance (no stimulation for ~1 month) before enabling therapy. It also documents substantial limitations in logging completeness (ECoG recordings ~3% complete versus Activity Logs 100%) and reports that extrapolated performance metrics significantly differ from standard methods (Wilcoxon $p < 10^{-6}$, >5% absolute mean differences). The excerpt does not provide quantitative efficiency, processing latency, or detailed algorithmic implementation (e.g., filter parameters, thresholds, computational cost)."

[6.1] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: "The paper evaluates a two-stage detection pipeline: a lightweight on-board gating detector (using a small feature bank including bandpower/spectral features, cross-correlation, and line-length) and a precise cloud-based algorithm. Bandpower (spectral) and line-length features were explicitly tested; adding cross-correlation and line length did not improve AUC over bandpower alone. A five-feature detector achieved 98–99% sensitivity across patients and reduced data transfer by 22.7 $\times$. The cloud precision detector identified all transmitted seizures with an average false positive rate of 0.03 detections/hour. Processing speed: the gating algorithm analyzed one hour of recorded data in 16.5 $\pm$ 0.3 s, and the precision algorithm analyzed one hour in 17.4 $\pm$ 0.2 s; when analyzing one-second clips the combined algorithms introduce latency under 10 ms. Raw iEEG is ~40 kb/s, within 4G streaming capability. Implementation considerations: authors limited number/type of on-board features to meet processing and battery constraints; performance plateaued after 2–3 features, making analog/on-chip computation feasible; spectral features are amenable to hardware implementation. The paper notes current devices (NeuroPace RNS) accept high false positive rates to reach sensitivity."

[6.2] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: "The excerpt describes a cloud-gating architecture that uses lightweight on-device gating to select candidate seizure clips for a cloud-based precision detector. Feature sets evaluated include bandpower (frequency) features, pairwise cross-

correlation between normalized channel power spectra (1–47 Hz), sorted eigenvalues of the cross-correlation matrix, and line length. A random forest of 3000 trees was trained on the union of three feature sets; both gating and precision algorithms were trained per-patient. Classification operated on one-second clips, with seizure likelihoods summed over a moving window (five seconds for gating, two seconds for precision) and an empirical threshold to generate detections. During gating tuning they applied class-weighted loss (interictal loss = 1; ictal loss = interictal/ictal ratio) to boost sensitivity. Frequency band (bandpower) features were most discriminative and “only two bandpower features is sufficient to maximize performance in this cohort.” Performance: a single-feature gating detector detected $\geq 92\%$ per subject (95% overall) and reduced iEEG transfer by 24.4 $\times$; a five-feature detector reached $\geq 98\%$ per subject (99% overall) with similar specificity and 22.7 $\times$ reduction. The cloud precision algorithm identified all transmitted seizures with an average false positive rate of 0.03 per hour. Latency was reported as compatible with real-time implementation.”

[6.3] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: “The paper describes a cloud-gating architecture where a lightweight on-device gating detector (using a small bank of device-compatible features such as bandpower, with optional cross-correlation and line-length features) selects data to stream to a more precise cloud-based classifier. A five-feature configuration yielded $\geq 98\%$ sensitivity per patient and 99% overall while reducing data transfer by a factor of 22.7. The cloud precision algorithm identified all transmitted seizures with an average false positive rate of 0.03 detections/hour. Computational latencies reported: the gating algorithm processed one hour of recorded data in 16.5 ± 0.3 s and the precision algorithm processed one hour of transmitted data in 17.4 ± 0.2 s; when analyzing one-second clips this corresponds to an end-to-end algorithmic latency under 10 ms. The authors intentionally limited on-board features for power/processing constraints, found performance plateaued after 2–3 features, and note spectral features are amenable to low-power hardware implementation. Raw iEEG bandwidth is ~ 40 kb/s, compatible with 4G streaming. A version has been implemented in a next-generation implantable device (Activa RC+S).”

[6.4] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: “The authors describe a cloud-gating architecture that uses an ultra-lightweight on-device gating detector to limit transmission to the cloud and reduce energy use and network dependence. The on-board detector’s performance “plateaued after including only 2 or 3 features,” making it feasible for analog on-chip processing; the device pre-processing is limited to data reduction and identification of potential seizure clips rather than signal transformation. Logistic regression was chosen “to ensure compatibility with device hardware” and “outperformed linear discriminate analysis (LDA), and performed as well or better than more complex methods (such as random forests and support vector machines) in the feature-limited environment.” Algorithm computation added “only trivial delay,” and with local off-board computation (belt device/phone) trials demonstrated “total latency (including data storage, transmission, analysis, and delivery of stimulation) under 200ms.” Bandwidth needs are low (“40kbps requirement for 16-channel iEEG streaming”). In 120,000+ hours, 14 of 982 seizures were undetected; some misses were due to a 1-feature detector’s sensitivity to morphology, while 5-feature misses were linked to recording/transmission gaps. The authors note limits of stepwise feature selection and emphasize need for on-board capability because continuous cloud-only analysis is currently impractical; they also highlight cybersecurity concerns for networked medical devices.”

[6.5] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: “The paper describes a two-stage cloud-gating architecture: a hypersensitive, lightweight on-device “gating algorithm” that flags potential seizure clips and a computationally intensive “precision algorithm” running in the cloud for final detection. The gating stage is implemented as a logistic regression trained on up to five simple, per-patient-selected features drawn from a small feature bank: bandpower features (Welch PSD across 25 manually-chosen,

partially-overlapping bands spanning 5–70 Hz), normalized pairwise cross-correlation (with 0.05 s lag), and line-length (sum of magnitude of first derivative). Features were selected via a greedy stepwise process minimizing cross-validation error. The gating design was co-developed with Medtronic to ensure compatibility with implantable hardware and was ported as a proof-of-concept to the Activa RC+S device. On-board computation in experiments used a MacBook Pro for simulation; cloud processing used a multi-core Xeon server. The precision algorithm operates on 1 s non-overlapping iEEG segments, extracts cross-correlation/eigenvalue, 1–47 Hz FFT power (normalized), and cross-correlation of normalized power spectra, and uses a 3000-tree random forest. The excerpt does not provide explicit latency or quantitative on-device performance metrics (sensitivity/false-positive rates) for real-time deployments."

[6.6] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: "The paper describes a cloud-gating architecture where a lightweight on-device gating algorithm continuously monitors implanted iEEG and identifies candidate seizure epochs for transmission to a cloud-based, computationally intensive detector. The on-device gate is explicitly characterized as "sensitive" and implemented with a "small feature set" feeding a linear classifier, making it compatible with next-generation implantable device computational constraints. Candidate epochs are sent to a cloud-based precision algorithm that leverages greater compute resources to perform accurate seizure detection. The approach was validated on eleven human NeuroVista implanted recordings. Reported performance for the combined (gate + cloud) system was 99% seizure detection sensitivity with a false positive rate of 0.03 per hour; the cloud precision stage "successfully identified all seizures transmitted by the gating algorithm while significantly reducing the false positive rate." The excerpt does not provide specifics on which features comprise the "small feature set," classifier parameters, per-epoch latency, transmission bandwidth/energy costs, or real-time runtime numbers on device, so those implementation details are not available in this excerpt."

[6.7] Cloud computing for seizure detection in implanted neural devices. Steven Baldassano, Xuelong Zhao, Benjamin Brinkmann, Vaclav Kremen, John Bernabei, Mark Cook, Timothy Denison, Gregory Worrell, Brian Litt. *Journal of Neural Engineering* (2019). <https://doi.org/10.1088/1741-2552/aaf92e> Context: "The excerpt describes a cloud-gating paradigm: a lightweight gating algorithm runs on-board an implant to selectively transmit data to a cloud analysis platform for up-front data reduction, allowing complex seizure detectors to run in the cloud. It notes that full-featured seizure algorithms typically use hundreds of spectrographic and cross-channel features and complex classifiers (random forests, neural networks) that are impractical on implants due to limited on-board computational resources and battery life. The paper cites existing on-device, low-complexity features/classifiers as practical examples: NeuroPace RNS uses signal line length, area, and halfwave features with empirically determined thresholds, and Medtronic Activa PC+S computes several bandpower features and can apply simple classifiers such as linear discriminant analysis (LDA). Telemetry constraints are emphasized: many devices can stream only four iEEG channels for roughly eight hours before recharge. Dataset and signal specs used in evaluation are given (intracranial EEG sampled at 400 Hz, lowpass filtered at 180 Hz; seizures annotated as earliest electrographic change). The excerpt does not provide explicit latency, on-device gating feature-set sizes, classifier runtimes, or quantitative gating/detection performance metrics, so those specifics are not extractable here."

[7.1] A hardware-algorithm co-design approach to optimize seizure detection algorithms for implantable applications. Shriram Raghunathan, Sumeet K. Gupta, Himanshu S. Markandeya, Kaushik Roy, Pedro P. Irazoqui. *Journal of Neuroscience Methods* (2010). <https://doi.org/10.1016/j.jneumeth.2010.08.008> Context: "The excerpt lists time-domain features evaluated for seizure detection in an implantable/real-time context: energy, coastline (line length), Hjorth variance parameter, RMS amplitude, rectified variance, non-linear autocorrelation and EEG power. Energy and RMS are defined via windowed squared samples/averages (e.g., $E[i] = x^2(i)$; Eavg computed over N-sample windows). Coastline/line-length is described as the sum of absolute differences between consecutive samples. Variance and rectified-variance are computed as windowed averages; non-linear autocorrelation is obtained via max/min operations across subwindows. The paper uses these features combined via a simple linear

combination to form a detection algorithm and cites prior use in implantable devices (Neuropace uses line length). The excerpt also contains rodent experimental details (twisted-pair 2-channel stainless steel electrode inserted into dentate gyrus, 3.5 mm ventral, insertion rates 100–300 mm/min), indicating in vivo model work. The excerpt does not provide explicit operation counts, memory-access counts, latency numbers, or quantitative performance metrics (sensitivity/specificity/latency) for the features or algorithms."

[8.1] A Review of EEG and MEG Epileptic Spike Detection Algorithms. Fathi E. Abd El-Samie, Turkey N. Alotaiby, Muhammad Imran Khalid, Saleh A. Alshebeili, Saeed A. Aldosari. IEEE Access (2018). <https://doi.org/10.1109/access.2018.2875487> Context: "The excerpt summarizes characteristics of interictal spikes and common automated detection approaches that are relevant to embedded real-time IED/LFP detection. Typical spike morphology: duration ~20–70 ms, amplitude >20 μ V ($\geq 1.5 \times$ background); MEG spikes show longer ranges (27–120 ms) with measurable slopes and curvature. Detection pipelines generally include three steps: pre-processing (SNR, artifact rejection), feature extraction, and classification. Most research is time-domain and feature-based: common features are amplitude, duration, slope, sharpness/curvature and half-wave measures. Example algorithms include Gotman's half-wave feature methods (reported 0.79 false positives/min after state classification) and DFAspike — a deterministic finite automata using only amplitude and slope with a 13-state diagram and a reported 99.13% recognition rate. The text notes artifact types (e.g., saccadic spikes) requiring filtering, and that automated methods still lag expert visual review. From an embedded/closed-loop perspective, simple time-domain feature extractors and DFA-like finite-state detectors are low-complexity, low-latency candidates; pre-processing for SNR and artifact rejection remains essential. The excerpt lacks explicit latency, computational-cost figures, and intracranial/LFP or animal-model-specific performance metrics."

[9.1] Closed-loop control of seizure activity via real-time seizure forecasting by reservoir neuromorphic computing. Maryam Sadeghi, Darío Fernández-Khatiboun, Yasser Rezaeiyan, Saima Rizwan, Alessandro Barcellona, Andrea Merello, Marco Crepaldi, Gabriella Panuccio, Farshad Moradi. Neuromorphic Computing and Engineering (2026). <https://doi.org/10.1088/2634-4386/ae3e14> Context: "The excerpt documents an FPGA-based neuromorphic seizure-control system (NCS) that uniquely combines seizure forecasting with forecast-driven stimulation, but notes prior systems "have not deployed real-time closed-loop stimulation." The authors validated anti-seizure efficacy in hippocampal spheroids (a reductionist in vitro model) and plan future validation in freely moving animals. A current NCS limitation is manual tuning of operating parameters; future work aims to add machine-learning auto-tuning. Relevant cited implementations and algorithm classes suitable for embedded real-time IED/interictal spike detection include: time–frequency feature extraction (STFT) with SVM classifiers, FPGA implementations of artificial neural networks and deep neural networks, low-power/approximate feature extraction for wearable prediction, spike-event neuromorphic processors with on-chip STDP, ReSuMe digital hardware for SNN learning, and specialized SNN cores (e.g., NEXUS). The excerpt itself does not provide explicit latency, computational complexity, or detailed performance metrics for intracranial EEG/LFP or animal-model IED detection; those details are likely in the cited references (refs. 30–34, 22, 23, 17, 24)."

[10.1] Time series segmentation for recognition of epileptiform patterns recorded via microelectrode arrays in vitro. Gabriel Galeote-Checa, Gabriella Panuccio, Angel Canal-Alonso, Bernabe Linares-Barranco, Teresa Serrano-Gotarredona. PLOS ONE (2025). <https://doi.org/10.1371/journal.pone.0309550> Context: "The paper describes two time-series segmentation algorithms (ZdensityRODE and AMPDE) for detecting epileptiform activity in LFPs recorded from 4AP-treated rodent brain slices via MEA. To keep computation light, the primary features used were event amplitude range and duration, though the authors note that higher-cost features (skewness, kurtosis, Shannon entropy) are possible. The approach explicitly aimed to reduce computational cost. Performance metrics (precision, recall, Jaccard, F1) are reported per session and averaged: ZdensityRODE: ictal P+ 93%, Re 93%, Jaq 86%, F1 91% (interictal P+ 42%, Re 45%); AMPDE: ictal P+ 96%, Re 90%, Jaq 86%, F1 92% (interictal P+ 54%, Re 33%). A multicriteria decision analysis weighted metrics by event type. Algorithm performance correlates with SNR (Pearson $r = 0.7$ training, $r = 0.4$ validation). The paper notes capability to detect interictal events as well as ictal, and references other works using SVM, ML, and discrete wavelet features

(kurtosis/skewness) with reported sensitivities/precisions. The excerpt does not provide low-level computational complexity (operation counts, memory accesses), latency, energy/implantable-hardware metrics, nor explicit closed-loop implant deployment details."

[10.2] Time series segmentation for recognition of epileptiform patterns recorded via microelectrode arrays in vitro. Gabriel Galeote-Checa, Gabriella Panuccio, Angel Canal-Alonso, Bernabe Linares-Barranco, Teresa Serrano-Gotarredona. PLOS ONE (2025). <https://doi.org/10.1371/journal.pone.0309550> Context: "The excerpt describes two time-series-segmentation (TSS) algorithms for low-power LFP-based detection: ZdensityRODE and AMPDE. ZdensityRODE achieved 93% precision and recall for ictal event detection and 42% precision for interictal events; AMPDE achieved 96% precision and 90% recall for ictal detection and 54% precision for interictal detection. The methods were validated on epileptiform activity induced by 4-aminopyridine in mouse hippocampus-cortex slices and recorded with microelectrode arrays (an in vitro rodent model). The paper frames these algorithms for use in implantable/closed-loop stimulation contexts, emphasizing "low power consumption through computationally inexpensive implementations" and noting embedded detection of biomarkers such as high-frequency oscillations (HFOs) for low-latency real-time seizure suppression. It also cites existing algorithmic approaches used in the field (PCA, linear SVM, approximate entropy, KNN, Naive Bayes, SVM-based feature extraction). The excerpt does not provide explicit computational-complexity details (operation counts, memory-access counts), latency numbers, specific feature implementations (e.g., line length, area, bandpower, wavelets, AR models), nor precise runtime/memory footprints; it notes the algorithms were initially trained for ictal detection but can be fine-tuned for interictal detection/prediction."

[10.3] Time series segmentation for recognition of epileptiform patterns recorded via microelectrode arrays in vitro. Gabriel Galeote-Checa, Gabriella Panuccio, Angel Canal-Alonso, Bernabe Linares-Barranco, Teresa Serrano-Gotarredona. PLOS ONE (2025). <https://doi.org/10.1371/journal.pone.0309550> Context: "The excerpt discusses energy-efficient, hardware-oriented real-time detection approaches and introduces two novel time-series segmentation algorithms (ZdensityRODE and AMPDE) developed to classify ictal and interictal events in LFP recordings. It surveys prior methods (phase synchronization, multilevel wavelet transform, Izhikevich networks, auto-ranging thresholding, Wiener algorithm) as more power-efficient than complex machine-learning classifiers, and stresses constraints from energy, memory, and simpler hardware implementations. Design principles highlighted: reduce computation, memory accesses, and power; minimize operation count; increase parallelism; and optimize hardware architecture. ZdensityRODE uses an adaptive z-score with short-term memory for faster computation and reduced operations; AMPDE uses scalogram-based peak detection and density estimation, providing better long-term adaptability at higher computational cost. Both apply cumulative look-forward time integration for post-processing and can distinguish ictal discharges, interictal events, and baseline. Experimental context: MEA recordings from mouse hippocampus-cortex slices with 4-AP (250 μ M), sampled at 2 kHz, are used to validate algorithms. The excerpt does not provide explicit numeric computational-complexity counts (ops or memory accesses), latency figures, or quantitative performance metrics (accuracy, sensitivity, latency) for closed-loop IED/interictal spike detection."

[11.1] Early Seizure Detection with an Energy-Efficient Convolutional Neural Network on an Implantable Microcontroller. Maria Hügler, Simon Heller, Manuel Watter, Manuel Blum, Farrokh Manzouri, Matthias DümpeImann, Andreas Schulze-Bonhage, Peter Woias, Joschka Boedecker. 2018 International Joint Conference on Neural Networks (IJCNN) (2018). <https://doi.org/10.48550/arxiv.1806.04549> Context: "The paper analyzes practical microcontroller implementation constraints for seizure-detection CNNs: they use 1 s input windows and STFT spectrogram preprocessing, and evaluate models with 1 s stride. Memory requirements include model parameters, a rolling-window preprocessing buffer, lookup tables, and two hidden-layer I/O buffers. SeizureNet achieves a strong trade-off (median sensitivity 0.96, median delay $\sim$ 3.7 s, fp/h 10.1 at threshold 0.5) and is implementable on their device; Kiral-Kornek's network is also theoretically implementable but its STFT preprocessing adds $\sim$ 38% runtime and produces high false positives (outliers up to 464/h). Acharya et al.'s large 16 s window yields low fp/h (median 0.04) but unacceptable delay (median 24.3 s) and low sensitivity (0.67); its fully-connected layers consume $\sim$ 74% of memory, preventing practical implementation. EEGNet and Acharya are infeasible due to high runtimes. A zero-mean approximation reduces runtime and memory but increases me-

dian delay to ~ 6 s. Power measurements: SeizureNet preprocessing+forward pass ≈ 850 μ W; an adapted TrueNorth-like implementation estimated at 4.3–7.5 mW (based on neuron counts). Classifier thresholding trades sensitivity vs false positives vs delay (higher thresholds reduce fp/h but raise delay and lower sensitivity)."

[11.2] Early Seizure Detection with an Energy-Efficient Convolutional Neural Network on an Implantable Microcontroller. Maria Hügler, Simon Heller, Manuel Watter, Manuel Blum, Farrokh Manzouri, Matthias DümpeImann, Andreas Schulze-Bonhage, Peter Woias, Joschka Boedecker. 2018 International Joint Conference on Neural Networks (IJCNN) (2018). <https://doi.org/10.48550/arxiv.1806.04549> Context: "This paper proposes SeizureNet, described as "the first convolutional neural network for seizure detection specifically for an implantable ultra-low power microcontroller," and reports detection performance of sensitivity 0.96, false detection rate 10.1/hour, and median detection delay 3.7 s. The target hardware is the MSP430FR5994 microcontroller; the excerpt lists its power characteristics as "power consumption of 118 μ A/MHz in active mode and 0.5 μ A in standby mode" and highlights on-chip "ferromagnetic nonvolatile memory (FRAM)" useful for fast low-power storage of network activations. The authors state the architecture "exhibits low runtime and memory usage" and that they "investigate approximations to comply with hardware limits while preserving accuracy." They also note comparisons to other small CNNs, an SVM, and neuromorphic approaches, and that they evaluate runtime, memory, and energy consumption relative to baselines. The excerpt emphasizes implant constraints (e.g., avoiding heating of surrounding tissue). Missing from this excerpt are concrete per-inference runtimes, exact memory footprint (bytes), quantization details (bitwidths), and energy-per-inference numbers — the paper claims low runtime/memory/energy but does not provide those numeric implementation details in the provided pages."

[12.1] A Closed-Loop Neuromodulation System Using Machine Learning for Real-Time Prediction and Suppression of Epileptic Seizures. PB Devi. 2025. <https://ijaiot.org/index.php/IJAIOT/article/download/2/2> Context: "The paper describes a lightweight 1D-CNN (input: 10-second multichannel iEEG) with three conv layers (filters 16, 32, 64; kernel size 3), batch norm/ReLU, global average pooling, dropout (rate=0.5), and a sigmoid output for preictal vs interictal classification. Training used Adam and binary cross-entropy with class weighting. For embedded deployment the model was optimized via quantization (32-bit float $\rightarrow$ 8-bit integers, "reducing the memory footprint by 75% with minimal accuracy loss"), pruning to create a sparse, lower-compute model, and conversion to TensorFlow Lite for efficient edge inference. Hardware work included a custom PCB and "an ultra-low-power ARM" microcontroller (model unspecified). Runtime in vivo used real-time analysis on a 2-second sliding window updated every 100 ms (on an external processing unit running the CNN). Reported performance: 89% seizure prediction (n=127/143) with 58.2 ± 12.4 s mean lead time, 0.15 false predictions per hour, and 85% suppression efficacy when stimulation was triggered. The excerpt does not provide specific MCU model, RAM/flash sizes, exact inference latency on MCU, clock/power numbers, or per-inference memory/runtime metrics."

[13.1] Edge deep learning for neural implants: a case study of seizure detection and prediction. Xilin Liu, Andrew G Richardson. Journal of Neural Engineering (2021). <https://doi.org/10.1088/1741-2552/abf473> Context: "This study implemented three DL models (DNN, CNN, LSTM) on a general-purpose off-the-shelf MCU for event-based seizure detection and prediction. Implementation notes: data-transfer time over USB was excluded from measured inference time; the CPU core was put to sleep after inference; power was measured directly from the supply and reported as the average within the segment period; reported memory sizes include code overhead for data handling. Runtime, memory and power (relative): DNN had the largest memory footprint but the shortest inference time and lowest power; CNN had the smallest memory (limited kernel size) and a balanced profile; LSTM gave the best detection performance but had the longest inference time and highest power consumption. Quantization reduced inference time, memory size, and power consumption without sacrificing performance. Computational cost: the CNN required ~ 2.4 M MACs to process a 256×9 input. Performance metrics (LOOCV): detection sensitivities/FPRs — DNN 87.36% / 0.169 h $^{-1}$, CNN 96.70% / 0.102 h $^{-1}$, LSTM 97.61% / 0.071 h $^{-1}$; prediction sensitivities/FPRs — DNN 76.66% / 0.710 h $^{-1}$, CNN 90.66% / 0.204 h $^{-1}$, LSTM 90.72% / 0.241 h $^{-1}$. The authors note potential improvements with intracranial EEG and ASIC or in-memory compute to further reduce power."

[13.2] Edge deep learning for neural implants: a case study of seizure detection and prediction. Xilin Liu, Andrew G Richardson. *Journal of Neural Engineering* (2021). <https://doi.org/10.1088/1741-2552/abf473>
Context: "The study implemented three DL models (DNN, CNN, LSTM) trained with TensorFlow on CHB-MIT scalp EEG and used a sliding-window weighted majority voting (WMV) detector. After iterative model compression and coefficient quantization, the authors deployed the models on a general-purpose off-the-shelf microcontroller for real-time testing and explicitly measured inference sensitivity, false positive rate (FPR), execution time, memory size, and power consumption. Reported seizure-event sensitivities/FPRs were: DNN 87.36% / 0.169 h⁻¹, CNN 96.70% / 0.102 h⁻¹, and LSTM 97.61% / 0.071 h⁻¹. Relative runtime/memory/power tradeoffs were noted: LSTM gave best detection at highest power, DNN had shortest execution time, and CNN offered balanced performance with minimum memory requirement. Model compression and quantization produced significant power and memory savings with <0.5% accuracy loss. These implementation details indicate feasibility of microcontroller-based edge DL for seizure detection, with quantization and compression enabling practical runtime, memory, and power tradeoffs for neural-implant or wearable deployments."

[13.3] Edge deep learning for neural implants: a case study of seizure detection and prediction. Xilin Liu, Andrew G Richardson. *Journal of Neural Engineering* (2021). <https://doi.org/10.1088/1741-2552/abf473>
Context: "The paper reports practical evidence relevant to cloud-gating and lightweight on-device gating for implantable, real-time detection. They implemented a cascaded approach: an edge DL model followed by a second-stage, sliding-window WMV algorithm to compensate for small input buffers. Quantization reduced inference time, memory, and power without hurting performance. Classifier trade-offs: DNN had shortest inference and lowest power but weaker prediction; CNN provided balanced performance with the smallest memory footprint; LSTM gave highest detection sensitivity and lowest FPR (CNN: sensitivity 96.7%, FPR 0.102; LSTM: sensitivity 97.61%, FPR 0.071) at higher latency and power. Hardware-aware metrics: their CNN needed ~2.4M MACs to process a 256×9 input matrix (implying the feature window is 256 samples × 9 channels), and layer size should be scaled with sampling rate to capture temporal features. For implantable use, they exclude external USB transfer time, put the CPU to sleep between inferences, and permit periodic (not continuous) upload to cloud for heavier analysis/model updates. These details support a lightweight, on-device gate (small CNN/DNN + sliding-window voting), with occasional cloud offload for retraining or deeper processing."

[14.1] A Comparison of Energy-Efficient Seizure Detectors for Implantable Neurostimulation Devices. Farrokh Manzouri, Marc Zöllin, Simon Schillinger, Matthias Dümpelmann, Ralf Mikut, Peter Woias, Laura Maria Comella, Andreas Schulze-Bonhage. *Frontiers in Neurology* (2022). <https://doi.org/10.3389/fneur.2021.703797>
Context: "This excerpt compares three patient-specific, energy-efficient real-time seizure detectors intended for implantable neurostimulation: random forest (RF), LSTM recurrent neural network (RNN), and convolutional neural network (CNN). The RF used 16 handcrafted time- and frequency-domain features as input, while CNN and RNN operated on raw EEG. Energy was estimated with a platform-independent model counting arithmetic operations (AOs) and memory accesses (MAs); estimates were validated by implementing the RNN on an ultra-low-power microcontroller. Reported median area under the precision-recall curve (AUPRC) scores were 0.49 (RNN), 0.47 (CNN), and 0.46 (RF). Per-classification operation and memory-access counts were: RF 67k AOs & 67k MAs, CNN 488k AOs & 963k MAs, RNN 772k AOs & 978k MAs, with MAs contributing more to energy. The paper notes seizure detection is feasible with few channels and highlights a problem of high false-detection burden in existing RNS devices (example derivation: mean stimulation period 5.9 min → >1,220 false detections per crisis event). The excerpt does not provide details on specific handcrafted features (e.g., line length, area, bandpower, wavelets, AR models), latency figures, nor mention interictal/IED detection or rodent-model closed-loop studies."